Justice (Hsiu-Yi) Lu

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Summary

Computational biologist interested in using data-driven and machine learning approaches to solve complex biological questions. Innovative and collaborative researcher with expertise in multi-omics integration, protein structure–function modeling, and scalable computational pipelines, with a track record of translating complex data into interpretable biological insights.

Work Experience

Graduate Student Researcher, Duke University, Hiroaki Matsunami Lab 2020 - Present

- Developed deep learning models in PyTorch to predict protein receptor–ligand interactions by integrating AlphaFold-predicted 3D structures, LLM embeddings, and transcriptomic data, uncovering features driving ligand selectivity.
- Performed model interpretability and feature attribution analyses to uncover the mechanistic basis of model predictions, identifying structural features that drive ligand selectivity for validated by guided protein engineering approach.
- Designed and implemented scalable single-cell, bulk RNA-seq, and proteomics pipelines to identify genetic regulators of protein expression, accelerating in vitro screening and target discovery.
- Built computational tools for visualization and exploration, geLine, an interactive web dash app for large-scale gene co-expression and maturation trajectory analysis, empowering researchers to explore developmental programs across cell types.
- Mentored trainees in both computational and experimental projects, fostering independent design and test skills.

ML Data Science Intern, Kriya Therapeutics 2023

- Engineered scalable ML pipelines for high-throughput drug target prediction, increasing 10× data throughput.
- Optimized data preprocessing and model training workflows using Python, scikit-learn, and cloud computing resources.
- Partnered with biology teams to incorporate omics-derived features into predictive models, improving biological relevance.

Research Technician II, Duke University, John Rawls Lab 2017 - 2019

- Conducted studies on host–microbe interactions in zebrafish, uncovering protective gut responses to pathogenic bacteria.

Awards

- 1st place Abstrax Prize: applications in taste/smell/analytical chemistry of 122 global teams, LLM Hackathon 2025
- Duke University IBIEM Computational Biology Fellow

Education

Duke University, Ph.D. in Molecular Genetics and Microbiology (Advisor: Hiroaki Matsunami) 2019 - 2025

- 1st place Duke LLM Hackathon 2025 | IBIEM Computational Biology Fellow | Taiwan-Duke Fellowship Recipient

Purdue University, B.S in Biological Science 2013 - 2016

Selected Publications (* First author)

- Identifying Key Regulators in Odorant Receptor Trafficking.
  Lu, HY*. & Matsunami, H. **Journal of Neuroscience** (2025)
- Physiology of olfactory sense.
  Lu, HY*., Meesa, P. & Matsunami, H., **Digital Technologies in Olfaction, 7-14** (2025)
- Role of iRhom2 in Olfactory Sensory Neurons: Implications for Odorant Receptor Regulation.
  Azzopardi, S*., **Lu, HY*.**, Monette, S., Rabinowitsch, A., Salmon, J., Matsunami, H., & Blobel, C. **IJMS** (2024)
- An odorant receptor that senses four classes of musk compounds
  Yoshikawa, K., Deguchi, J., Jieying, H., **Lu, HY**., & Matsunami, H. **Current Biology** (2022).
- Enteroendocrine cells sense bacterial tryptophan catabolites to activate enteric and vagal neuronal pathways.
  Ye L, Bae M, Cassilly CD, Jabba SV, Thorpe DW, Martin AM, **Lu HY.**, … & Rawls JF. **Cell Host Microbe** (2021).

Skills

Computer Programming Language: Python, R, Bash, Git
Bioinformatics: omics analysis (single-cell/bulk RNA-seq, proteomics), protein modeling, Alphafold, ESM, Boltz, Dash
ML & Data Science: PyTorch, CNNs, regression/classification, scikit-learn, RDKit, Pandas, NumPy, Scikit-learn, SciPy
Interest: Building pipelines and tools for optimization. Using data driven approach to solve complex biological questions